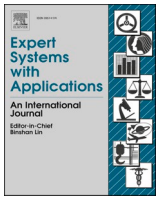
Expert Systems With Applications 229 (2023) 120492



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa



Review



A survey on location and motion tracking technologies, methodologies and applications in precision sports

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ARTICLE INFO

Keywords:
Sensor
Motion tracking
Data analytics
Precision sports
Data privacy

ABSTRACT

Sports involve commonly players and equipment of high dynamics. Their location and motion data are essential for sports digitalization-related applications, such as from monitoring player skill level to even media presentation of sports. Location and motion data of players and equipment are important elements for precision sport analytics. With the popularization of the Internet of things, a variety of sports location and motion tracking technologies have advanced in terms of sensors and tracking methodologies to achieve ubiquitous coverage from outdoors to indoors, higher precision, and applicability for increasing sports scenarios. This paper presents a systematic survey on location and motion tracking technologies that have been applied in sports to drive various sports-related analytics and applications. First, several wearable sensor-based technologies are introduced with their principles, advantages and shortages in sports, and the location and motion tracking methods of each technology. Then, vison-based technology and its methodologies that detect, track, and recognize objects including human and equipment in sports are illustrated. Furthermore, sports-related data analytics and applications are presented using location and motion data of human and equipment. Data privacy and ethics are also focused based on the location and motion tracking technologies. Finally, challenges and further development of location and motion tracking in precision sports are discussed. The comprehensive survey provides a survey about state-of-the-art of location and motion sensors and methodologies, as well future trends of these technologies, which are beneficial for developing forthcoming precision sports.

1. Introduction

Precision sports are committed to exploring better training and performance analysis methods. Location and motion data are two fundamental elements of precision sports, which digitize the movement process to analyze the technical level and performance of athletes. Precision sports analytics can evaluate the skills, strengths, and weaknesses of players in training and competitions by quantifying their posture and movement. Based on these quantitative indicators, coaches can develop personalized training plans to maximize the potential of athletes and improve their technical level (Yu, 2020).

Location and motion tracking technologies provide a series of intelligent solutions for professional sports. These technologies have been incorporated with intelligent sports equipment, such as shoes, insoles, socks, and underwear. Traditionally, the performance of players was primarily analyzed qualitatively by capturing images and video highlights. Traditional motion analysis methods are limited by the inability to obtain detailed location and motion tracking data, but now it is possible to quantify performance indicators and movement parameters to analyze the technique level and physical status of players based on various sensor technologies, even in high-dynamic sports, such as short track speed skating. In addition to performance evaluation, sensors embedded in sports equipment enable coaches to monitor the rehabilitation and recovery of players from injuries. Furthermore, precise sports digitalization makes the media display more diversified and greatly enhances the viewing experience of audiences by combining visualization technologies with sensor technologies. Location and motion tracking technologies have changed the traditional mode of sports analysis and accelerated the generation of precision sports, which revolutionize the training method of players and the guiding method of

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https://doi.org/10.1016/j.eswa.2023.120492
Received 21 September 2022; Received in revised form 31 March 2023; Accepted 12 May 2023
Available online 18 May 2023
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coaches.

Two types of athlete data are typically collected, namely, location/tracking data and biometric data. Position/tracking metrics measure the actual position of a player or other sports instrument. Such data includes position, acceleration, lateral movement, speed, and jump height, which are collected through video analytics or sensors coupled with global positioning systems and ground-based wireless networks. Biometric data refers to any biological information from a player. Measurements can include pulse rate, blood sugar or blood oxygen levels, sweat rate, sleep rhythm, and more. This article will focus on location/tracking data-related technologies, methods, and applications.

Precision sports based on accurate location and motion tracking is an emerging field integrating information and communications technology, spatial data (mainly location and motion) and sports science. However, some challenges have yet to be addressed for precision sports development. Therefore, this systematic survey provides location/motion technology researchers and precision sports designers with a survey about the state-of-the-art wearable sensors and vision technologies, methodologies, and data analytics-based applications. Challenges and future trends are also discussed regarding these three aspects. The structure pyramid of this article is shown in Fig. 1.

- Wearable sensors applicable for sports location and motion tracking are reviewed, including Global Navigation Satellite System (GNSS), Inertial Measurement Units (IMU), cameras, Ultra-wideband (UWB), Radio Frequency Identification (RFID) and Bluetooth Low Energy (BLE). Based on these sensors, their location and motion tracking and recognition methods used in sports are illustrated.
- Besides wearable sensors, vision-based location and motion tracking technology and its methodologies applicable for players and sport equipment are presented, including object detection, tracking, recognition, and classification.
- Furthermore, data analytics-based sports applications are explored using the location and motion data for professional sports.
- Athlete privacy risks and sports ethics issues were discussed in detail, and privacy enhanced computing technology for data protection was introduced.
- Challenges and future trends are discussed from the sensor technologies, methodologies with semantic information development, the future personalized sports and immersive visualization and ethic and personal privacy.

Reviews over the past few years have generally focused on one or multiple specific technologies applied in a particular sport group, as shown in Table 1. In contrast, this paper presents a more comprehensive introduction to location and motion technologies in sports, and emphasizes the importance of data privacy in the era of big data. It is expected that the systematic review on the location and motion technologies and methodologies applied in sports provides a guidance and reference for precision sports researches.

Fig. 2 presents the main structure of Sections 2–5. Section 2 introduces the main location and motion tracking sensors in sports. Based on sensors, Section 3 illustrates the methodologies of location and motion tracking in sports, and precision sports analysis based on location and motion tracking is presented in Section 4. While discussing technology applications, we also elaborate on data privacy protection and security in Section 5. Then, a discussion of challenges and trends in precision sports concerning the location and motion tracking technologies, methodologies and sports applications, data protection and legislation enhancement is explored in Section 6. Finally, we conclude this paper in Section 7.

2. Wearable sensor-based location and motion tracking technologies in sports

State-of-the-art sensor technologies provide accurate location and motion tracking data to digitalize sports. In this section, various

Table 1
Previous survey about sensor technologies in sports.

Ref.	Technology	Sport Genres
Cummins et al., 2013	GPS, IMU sensors	Team sports
Shih, 2017	Video	All sports
Camomilla et al., 2018	Wearable sensors	All sports
Shepherd et al., 2018	IMU sensors	Wheelchair Court Sports
Kamble et al., 2019b	Video	Sports with balls
Worsey et al., 2019	IMU sensors	Combat sports
Aroganam et al., 2019	Wearable sensors	Consumer sports
Rana and Mittal, 2021	Wearable sensors	All sports

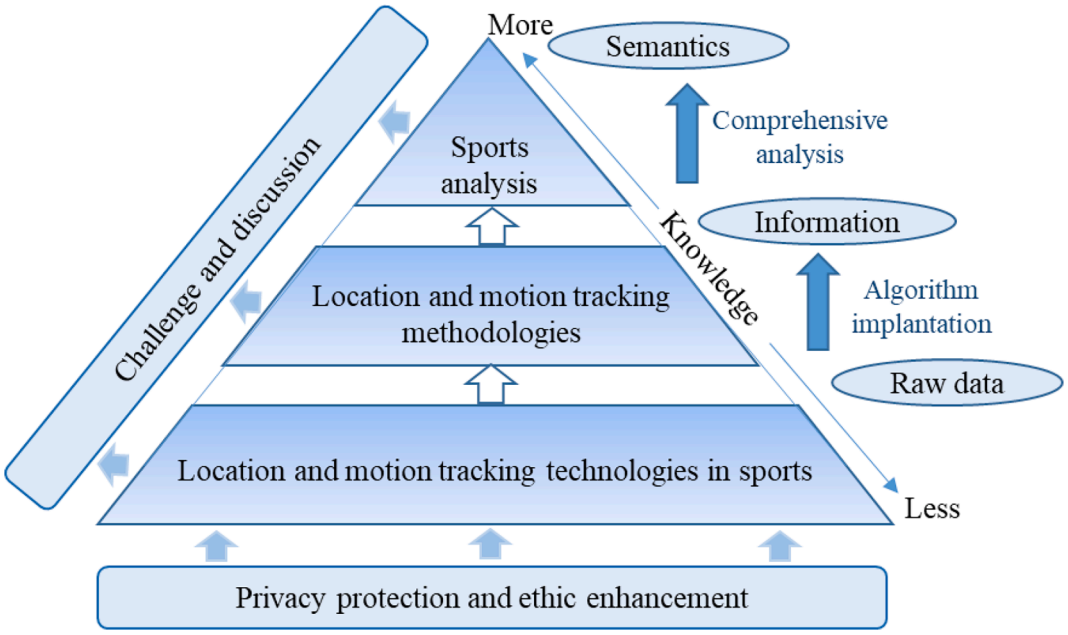


Fig. 1. The main structure pyramid of the article.

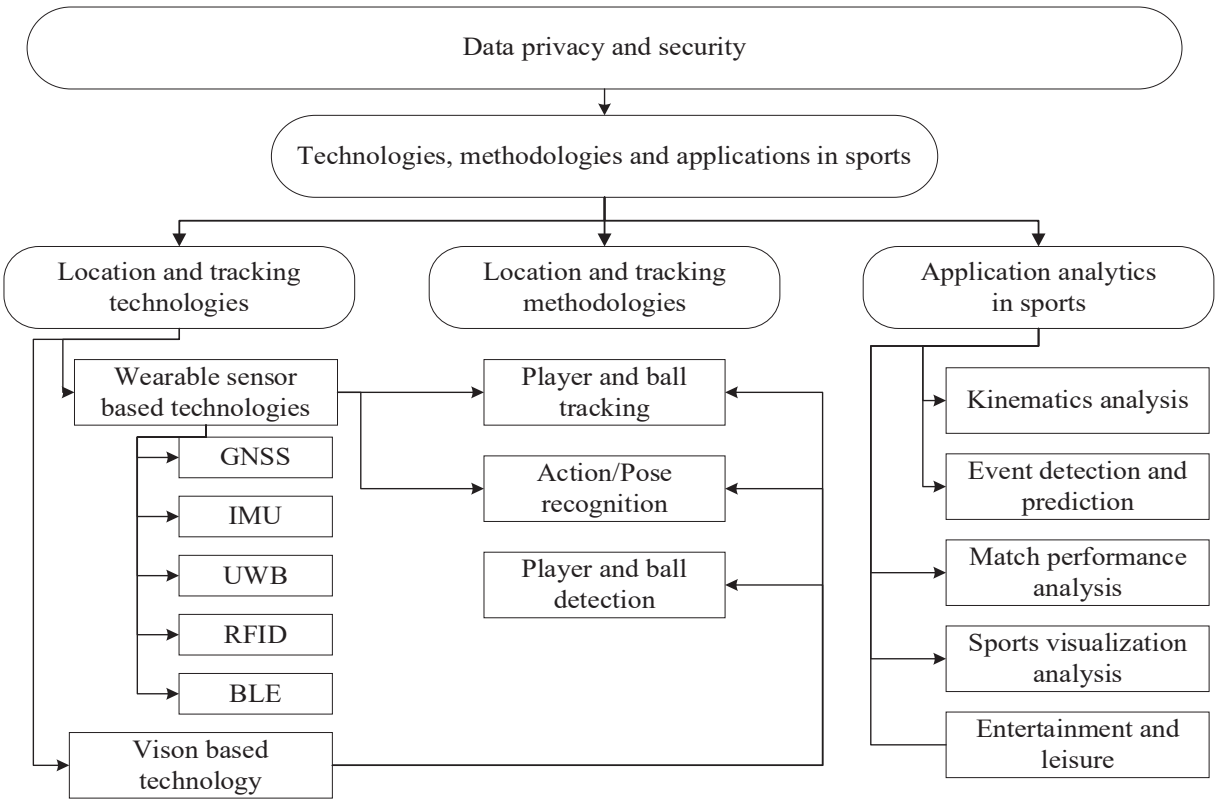


Fig. 2. Main contents of Sections 2–5.

wearable sensor technologies used in sports and their tracking and motion recognition methods are introduced thoroughly.

2.1. *Wearable sensor-based technologies*

Wearable sensor-based location and motion tracking technologies mainly involve GNSS, IMU, UWB, RFID and BLE (Liu et al., 2012). Generally, the characteristics of different technologies can affect their application performance in sports. Table 2 compares the advantages and disadvantages of the sensor technologies used in sport movement tracking and location.

Global Navigation Satellite System. GNSS is widely used in various outdoor environments since it can provide all-weather global positioning services. Through distance intersection, the position of the GNSS receiver can be obtained by measuring the distance between it and four or more satellites with known positions. Portable GNSS units, which are generally installed on sports equipment and clothing, extends their application to sports, which enables the location and tracking of

Table 2
Comparison of location techniques in sports: typical performance and application.

Sensor	Accuracy	Sampling rate	Suitable sports	Factors affecting accuracy	Advantages	Disadvantages
GNSS	mm ~ m	1 Hz, 5 Hz	Long and moderate distance linear running in outdoor sports	Sampling rate	Global absolute position, high precision when converged (with RTK/PPP)	Signal degradation in urban valley and indoor environment, weak in attitude determination
		10 Hz	Fast movements across short distances running in outdoor sports			
		15 Hz+	Fast movements across short distances running in outdoor sports			
IMU	Depend on the characters of sensors	100 ~ 1000 Hz	Gait analysis in race walking, collision detection in team sports, movement recognition and analysis in individual sports, team sports, water sports and snow sports	Attachment position	Self-contained, Robust in the short term, off-the-shelf sensors in consumer devices, full motion states	Relative navigation solution, high cost for high-end sensors, Significant errors for low-cost sensors
UWB	cm ~ m	8 ~ 20 Hz	Indoor sports positioning and tracking	Tag positions, Height of antennas	High-precision ranging, immune to inference, cost is reducing, being used in consumer devices	Require extra infrastructure, inherent to wireless signals, shorter range
RFID	dm ~ m	/	Circle-counting and timing in long distance sports, pressure measurement checks on team sports balls, balls and players detection and tracking in ball games	Density of RFID tags, Structure of antennas	Remote reading, Pollution resistance and durability	High cost, safety and privacy problem, low localization accuracy, product incompatibility
BLE AOA	dm ~ m	1 ~ 2 s	Players positioning and tracking	/	High-precision	High cost

individual or group movements.

The GNSS sampling rate is crucial in the reliability and validity of measurements. The explored sampling rates mainly include 1 Hz (Coutts and Duffield, 2010), 5 Hz (Portas et al., 2010), 10 Hz (Vickery et al., 2014) and 15 Hz (Johnston et al., 2014). Considering both positioning accuracy and device cost, it has been proven that 10 Hz is the best sampling rate when applied for tracking and positioning in sports, and it seems that there are no obvious additional benefits with a higher rate, which may be detrimental to the accuracy of distance and velocity (Varley et al., 2012).

Inertial Measurement Units. The IMU is the sensor unit containing inertial sensors that measure acceleration, angular acceleration, and sometimes magnetic field. The accelerometer measures accelerations along three axes, and then the velocity can be obtained through integration. The gyroscope measures angular accelerations, and then the attitude can be determined. The magnetometer measures magnetic field strength on three axes. Recent developments in microelectromechanical systems (MEMSs) have revolutionized to take advantage of the small size and light weight of MEMS IMU sensors for motion analysis in sports by embedding inertial sensors into wearable devices or binding sensors directly to the player's body or sports equipment. It overcomes the problems of limited vision, limited distance, and limited time delay in other motion analysis systems (Townshend et al., 2008). Furthermore, the unobtrusive and lightweight nature of the IMU sensor, with direct contact with the object, enables reliable data acquisition in challenging conditions, such as underwater sports.

Although the IMU data update frequency is high enough to reduce information loss and its small size and portability do not interfere much with players or coaches, the gyroscope and accelerometer deviate due to changes in physical characteristics. Meanwhile, when the measurement signal is integrated once or twice, the deviation accumulates linearly or quadratically with time. The IMU is installed on different joints of the player's body and different sports equipment, so the measurement accuracy of the IMU depends on its attachment position to the body, as shown in Fig. 3. In addition, indoor magnetic interference can lead to direction estimation errors, such as a water pump at the bottom of a swimming pool.

Ultrawideband. UWB is a wireless carrier communication technology that uses nanosecond nonsinusoidal narrow pulses to transmit data. It uses a frequency bandwidth of 1 GHz or more, and it occupies a large spectrum. Because UWB technology is insensitive to channel fading and the positioning accuracy can reach the centimeter level, high-precision positioning indoors and outdoors can be achieved, especially for indoor conditions, where the GNSS signal is not available, and other radio frequency (RF) technologies have problems in accuracy. UWB can be used for localization by using the time of flight (TOF) or time difference of arrival (TDOA) of the signals to obtain the distance of the target and the reference points. Generally, reference anchor nodes with known coordinates are arranged in sports venues, and the placement positions of the on-body tag are shown in Fig. 4.

Due to the low complexity of the UWB system and its centimeter-level positioning accuracy, it is a promising technology for motion analysis in sports. The performances of ten different UWB localization systems are compared in (Zwirello et al., 2014), which shows that the root mean square error of the positioning result is between 7 cm and 18 cm in small scenarios. However, considering the high dynamic activities in some sports, the position of the on-body tag, height of anchor nodes and the tag, and movement patterns of players can influence the positioning accuracy and communication reliability due to body occlusion and unpredictable movements.

Radio Frequency Identification. RFID is an automatic identification technology using electromagnetic transmission to perform noncontact two-way data communication. An RFID system contains a reader that can identify and communicate with RFID tags via RF signals. When a tag enters the RF signal coverage area, it can either send a message after receiving the RF signal from the reader or actively send a signal at a certain frequency. Then, the information is sent to the central process system for data analysis after being decoded by the reader using a predefined RF and protocol.

Bluetooth Low Energy. Bluetooth (or IEEE 802.15.1) is a short-range wireless technology that can transmit information between fixed or moving devices in a certain space. BLE, which is the low power variation of the original traditional Bluetooth standard, can be used for localization with different localization principles, such as received signal strength indication (RSSI) and angle of arrival (AOA). BLE AOA uses a single antenna to transmit a direction-finding signal. When the antenna array of the receiving device receives the signal, a phase difference is generated, and the relative signal direction can be calculated to obtain the position of the tag. Compared to the low accuracy of 5 m–10 m of RSSI-based BLE, BLE AOA can reach consistent submeter accuracy, which can be used in sports, even high-speed and highly dynamic sports. Moreover, BLE AOA costs less and provides high accuracy comparable to

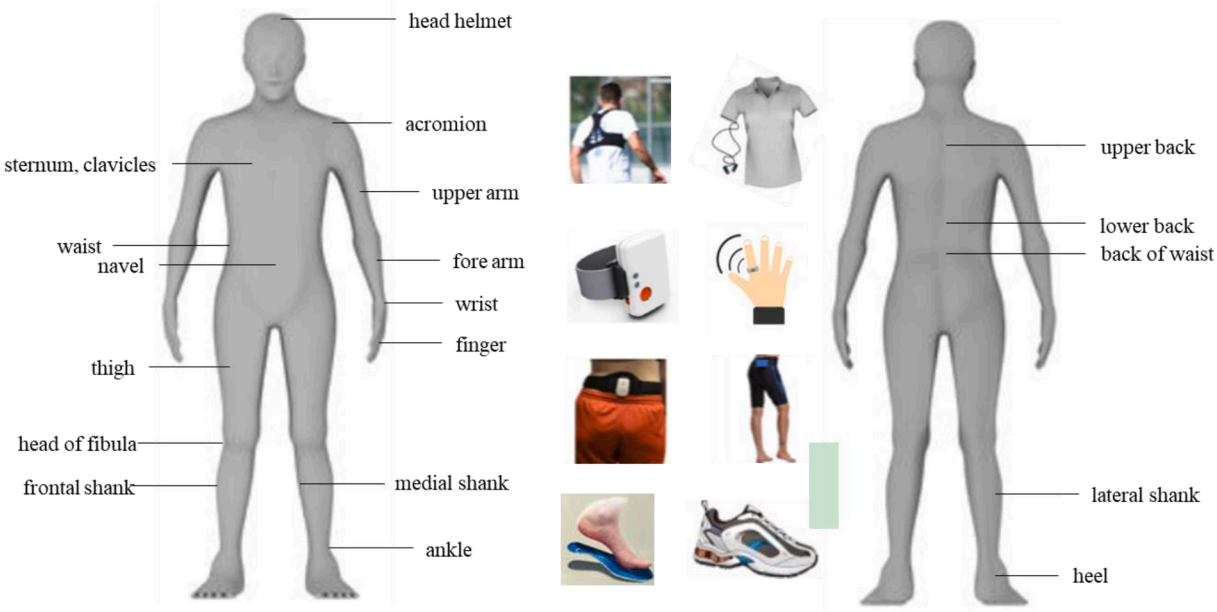


Fig. 3. The most common IMU devices attachment position and fixing method.

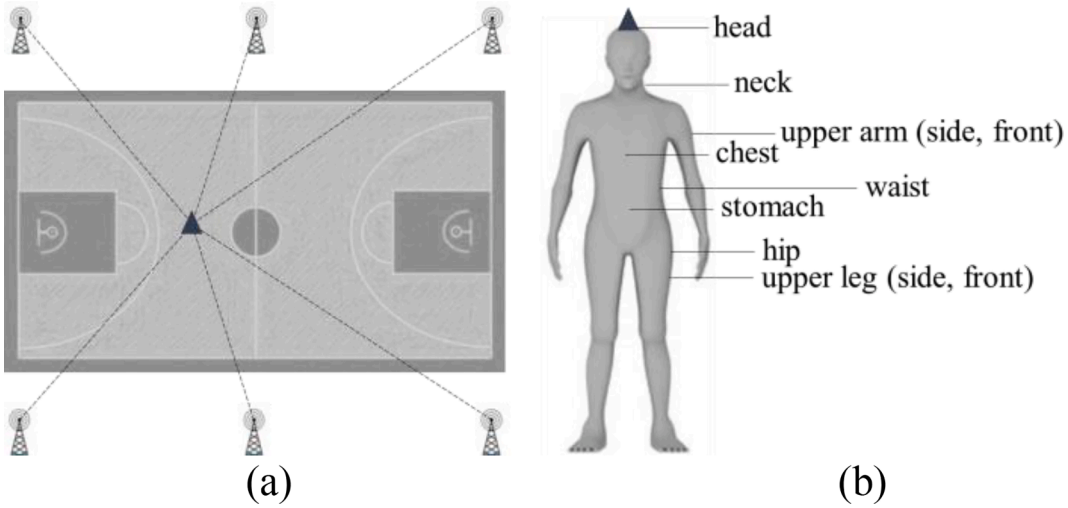


Fig. 4. (a) UWB positioning and (b) common positions of tags.

UWB-based solutions but at lower cost.

2.2. Wearable sensor-based tracking

Wearable sensor-based systems have been widely applied in a variety of sports for trajectory tracking, such as elite hockey (Jackson et al., 2018), cross-country skiing (Swarén et al., 2016), endurance racing (Gløersen et al., 2018), alpine sports (Samuelsson et al., 2016), basketball (Leser et al., 2014) and cycling (Minne et al., 2019).

GNSS has better results in sports with simple trajectories, such as rugby and football, but has limited research in high dynamics sports with complex movements, such as netball, lacrosse, and hockey (Ishii et al., 2002). A variety of basic components of movements can be measured by GNSS, including distance covered, velocity, accelerations, and decelerations. By analyzing the above basic measurements, physical load, athletic level, competition performance and training intensity can be quantified objectively.

IMU sensors focus on performance monitoring and motion recognition (Camomilla et al., 2018). Measurement analysis refers to detecting a variety of parameters measured by IMU sensors, including acceleration amplitude, instantaneous and average velocity, horizontal displacement, and vertical displacement, as well as joint movement direction and angular displacement. IMU-based motion pattern recognition is generally further quantified and analyzed through the abovementioned measurement analysis because different patterns have specific measurement statistics.

UWB localization is available in a variety of indoor and outdoor sports for motion tracking and analysis of players (Chantaweesomboon et al., 2016; Ridolfi et al., 2018; Bastida-Castillo et al., 2019). The ideal placement of the tag on the body is the head with minimum information loss and maximum anchor node visibility. Moreover, the UWB tag can also be installed on the sports equipment for its motion analysis, such as the trajectory analysis of the cricket ball, the hitting position analysis of the tennis ball (Gowda et al., 2017) and the position of wheelchair in wheelchair court sports (Rhodes et al., 2014). Many companies have produced commercial UWB systems for location and motion tracking in sports (Jiménez and Seco, 2016; Ruiz and Granja, 2017).

RFID technology is applied to personnel tracking in long-distance sports such as marathon running, road racing, skiing, and international cycling (Paek et al., 2013). The RFID tag embedded in the sports clothing communicates with the reader across the start and finish lines. RFID-based timing and management systems are widely used in track and field applications (Andersons and Ritter, 2015), and they can display real-time competition information, such as running laps and total running lap number (Satitsuksanoh et al., 2017). The National Football League is the first major league to track player movements by placing RFID tags on a player's shoes or clothing. RFID technology is also used as an alarm for blind players in Paralympic track and field sports (Almeida and Pinheiro, 2014). This system uses active tags fixed on the track surface to provide a player's actual position on the track. Furthermore, RFID can be used in ball games to detect the position and speed of the ball, including golf, hockey, volleyball, football, and basketball (Botwicz et al., 2017). RFID is also used for pressure detection on a ball since defects and pressure loss can affect the bounce and trajectory of the ball and has been applied in team sports such as football, handball, volleyball, and basketball (Rennane et al., 2018).

BLE AOA technology is available for all sports due to its high accuracy. For example, in fast-moving sports such as ice hockey and basketball, the position of players on the court can be obtained in real time, and the overall level of the team can be analyzed and evaluated during and after the game. It is even possible to transfer positioning from indoor to outdoor. In the case of golf, by positioning the golf ball with a built-in tag, the flight trajectory of the golf ball can be obtained, and the level of the shot can also be analyzed and evaluated.

In addition to using a single technology only, integration systems of different positioning technologies are studied for higher accuracy and robustness. For example, the inertial navigation system is integrated with GNSS to determine the trajectory of players in skiing, snowboarding and mountain biking (Sadi and Klukas, 2013). Map (Hallmén, 2015) and videos (Feuerhake et al., 2015) are also used as constraints. Furthermore, UWB and IMU sensors are applied for the trajectory of players in tennis (Umek et al., 2018). For device-based ball tracking, RFID technology is also used to measure the distance the ball moves. Moreover, multiple sensors are integrated to improve the accuracy, such as the integration of IMU and UWB, which are used for 3D trajectory tracking in cricket balls (Gowda et al., 2017).

2.3. Wearable sensor-based action recognition

Wearable sensor-based action recognition measures signals related to player movements, and sensors are often bounded on a player's wrist, hip, shoulder and arms. Data collected from sensors are analyzed by various algorithms, such as statistical characteristics analysis and HMM, for serve type recognition of beach volleyball players (Cuspinera et al., 2016), swing movement recognition of baseball players (Ghasemzadeh and Jafari, 2011) and swimming stroke recognition (Wang et al., 2019). Some activities are identified by measuring body segments in relative or absolute orientation. For example, the orientation of the lower-body

segments (sacrum, thighs, shanks) and skis were measured during the entire jump sequence in ski jumping (Chardonnens et al., 2013). Moreover, measurement statistics have been used to quantify the aerial acrobatics in snowboarding (Harding et al., 2008) and to evaluate the severity of collisions and tackles in rugby (Gabbett et al., 2011) and Australian football (Gastin et al., 2013). In addition to traditional methods, deep convolutional neural networks with sensor signals as input have also been studied to recognize actions in various sports, such as beach volleyball (Kautz et al., 2017) and cross-country skiing (Marsland et al., 2012). Table 3 lists different wearble sensors used for object tracking and recognition in different sport events.

3. Vision-based location and motion tracking technologies in sports

As a noncontact sensing technology, vision-based location and motion tracking uses images or videos captured by cameras for sports analysis. Traditional sports analysis is mainly based on experience, eye observation and manual recording (Duthie et al., 2003). However, its reliability depends on the number of observers, their experience, and the quality of their perspective. The development of computer vision makes it possible to capture movements of the player and sports equipment in a noncontact way. By analyzing the spatial and temporal relevance among video frames, information that players and coaches are interested in can be obtained for scientific training guidance. Recently, active 3D sensing has also been applied, especially for real-time monitoring in sports (Sasaki et al., 2018).

Sports video analysis mainly includes edited broadcast videos and nonedited videos. Edited broadcast videos contain replays and changes of perspective to the match. Nonedited videos are captured by single or multiple fixed cameras. A single camera is often fixed to detect movement taking place in parts of large venues, such as soccer (Kasuya et al., 2010). In addition, one camera is sufficient for some sports, such as Ping pong (Liu et al., 2014). Multiple cameras installed in different fixed positions are basically used for covering the whole sports court (Conaire et al., 2009), and camera calibration is necessary. Vison-based methods rely on the environment and are susceptible to weather and lighting conditions, so the performance of vision technology degrades in harsher conditions, such as darkness and fog. Due to the interference of light changes, it may be difficult to distinguish the foreground (i.e., the target object) from the background. Furthermore, object tracking can be achieved using a single tracker, where each player uses one tracker. However, this single target tracking method lacks a global description of the scene, so it is unable to track the target when the player is blocked. Another way is to track multiple targets frame by frame. In team sports, players may also be partially or completely occluded, resulting in tracking failure because of multiple moving objects and different camera angles. Table 4 lists different vision-based location and motion tracking methodologies for object detection, tracking and recognition in various sports.

3.1. Video- based player and ball detection

The player and the ball are the main objects in sports since their movements are related to key sports events, so proper object detection is the basis of sports analysis. Vision-based player and ball detection has been widely studied and can be divided into traditional methods and deep learning methods.

Traditional methods of object detection mainly include background subtraction (Zivkovic et al., 2003), frame difference (Ramya and Rajeswari, 2016) and optical flow (Ayed et al., 2015). Background subtraction can be divided into nonadaptive and adaptive methods by comparing the current frame with the reference background frame. Adaptive methods include Kalman filters, Gaussian mixture models (Gunes and Guldogan, 2016) and hidden Markov models (Balaji et al., 2020). The median filter, mean filter and pixelwise voting are nonadaptive methods, which have the drawback of great complexity. Background subtraction is less robust because it is sensitive to the environment. Frame difference can be regarded as a threshold setting method with simple computation. By calculating the difference in the corresponding pixel values between adjacent frames, the foreground and the background are separated and converted into a binary image according to the threshold value. However, both background subtraction and frame difference approaches may be useless if the background is moving. In sport broadcast videos, a match is usually the montage captured by multiple moving cameras. In this case, the background is constantly changing and moving, which is different from fixed cameras. The optical flow method is suitable for fixed and moving cameras, which detect objects by comparing the difference in motion between the target and the background.

Player detection includes single player detection and multiple player detection. For player detection in single-object sports, low-level appearance features, such as texture, color and shape, are used as handcraft representations by identifying key frames in downhill skiing, golf, bowling, ski jump and baseball games (Luo et al., 2003; Chen et al., 2011). To increase the accuracy of player detection in complex environments with different light conditions and various camera view changes, sports scene information is also used as experienced knowledge

Table 3
Comparison of wearable sensor-based location and motion tracking methodologies in various sport events.

Genres	Focus	Data Source	Methodology in Ref.	Sport events
Wearable sensor-based object tracking	Player tracking	GNSS	Swarén et al., 2016; Samuelsson et al., 2016; Gløersen et al., 2018	Cross-country skiing, Endurance racing, alpine sports
	Player tracking	UWB	Leser et al., 2014; Minne et al.,2019	Basketball, cycling
	Player's jump trajectory tracking	GNSS + INS	Sadi and Klukas, 2013	Skiing, snowboarding, mountain biking
	Player tracking	GNSS + Map	Hallmén, 2015	
	Player tracking	GNSS + Video	HMM and the Viterbi algorithm (Feuerhake et al. 2015)	Soccer
	Player tracking	IMU + UWB	Umek et al. 2018	Tennis
	Ball tracking	IMU + UWB	Gowda et al. 2017	Cricket ball
Wearable sensor-based action recognition	Serve types	Gyroscope	A warping longest common subsequence algorithm (Cuspinera et al., 2016)	Beach volleyball
	Skiing movement patterns	GNSS, accelerometer, gyroscope	Statistic characteristics analysis of different movements (Marsland et al., 2012)	Cross-country skiing
	Swing movements	Accelerometer, gyroscope magnetometer	Coordination analysis and template matching (Ghasemzadeh and Jafari, 2011)	Baseball
	Swimming style	IMU	HMM (Wang et al., 2019)	Swimming
	Recognizing ten actions	Accelerometer	Deep convolutional neural network (Kautz et al., 2017)	Beach volleyball

Table 4
Comparison of vison-based location and motion tracking methodologies in various sport events.

Genres	Focus	Data source	Methodology in Ref.	Sport events
Object detection	Player detection	Video	Metaheuristic algorithms (Balaji et al., 2020),A cascaded CNN (Luo et al., 2003),Dynamic Bayesian networks (Chen et al., 2011),Faster R-CNN (Bu, et al., 2017),Mask R-CNN and spatiotemporal interest points (Pobar and Ivasic-Kos, 2019),Region Proposal Network and R-CNN (Liu and Bhanu, 2019)	Volleyball, Soccer, Basketball, Downhill skiing, Golf swing, Bowling, and Ski jump, Roller skating, Handball
	Player detection	Broadcast video	Cubic B-spline interpolation (Jiang et al., 2009),Non-dominant color extraction filter and edge detection filter (Hamid et al., 2010)	Baseball, Tennis
	Player detection	Fixed cameras	Particle filter for detection after background and shadow removal (Direkoğlu et al., 2018),A particular diffusion equation and Support Vector Machine (Wang et al., 2018),CNN (Şah and Direkoğlu, 2018)	Soccer, Field hockey
	Ball detection	Video	The attention mechanism and faster R-CNN (Chakraborty and Meher, 2013),Appearance information (Leo et al., 2013),A probabilistic method of the local affine invariant region (Almajai et al., 2012),Background subtraction and appearance information (Kang et al., 2018)	Football, Basketball, Soccer, Tennis
	Ball detection	Broadcast video	A directed weighted graph (Yu et al., 2004)	Soccer
Video-based object tracking	Ball tracking	Video	Dynamic Kalman filter (Kim and Kim, 2009),Two-layered data association (Zhou et al., 2015),A weighted graph (Yu et al., 2007),A conditional random field-based model (Wang et al., 2014),Considering spatio-temporal relationships (Miura et al., 2009),RNN (Shah and Romijnders, 2016),Pre-trained CNN (Kamble et al., 2019a)	Soccer, Tennis, Basketball
	Ball tracking	Broadcast video	Enhanced particle filter (Yang and Li, 2017)	Soccer
	Player and ball tracking	Video	A constrained Bayesian inference model (Wang et al., 2016)	Basketball, soccer
	Player tracking	Video	A hierarchical data association method (Liu et al., 2013), A hierarchical model based on context knowledge and unscented Kalman filter (Liu et al., 2009),Spatiotemporal context learning (Zhang et al., 2018),The boosted particle filter (Lu et al., 2009)	Basketball, hockey, Short track speed skating, Soccer, Hockey
Vison-based action recognition	Actions including serve, dig, pass, spike, block, save, walk, and run in Beach volleyball Motions including run, walk, hop, jump, unknown and shot recognition including forehand, backhand, smash, and unknown in badminton	Video	A long-term recurrent region-guided convolutional network (Kong et al. 2020)	Beach volleyball, badminton
	Players' movements including skating down, up, left, and right	Video	HOG descriptors incorporated with the Sparse Multinomial Logistic Regression classifier (Lu et al., 2009)	Hockey
	Jump actions	Video	Pulse-like tracks and a physics-based validation (Chen et al., 2013)	Volleyball
	Player's actions	Video	SVM classifier with weighted trajectory-aligned descriptors (Yu et al., 2014),SVM classifier with dense trajectory of interest points (Abdul-Azim and Hemayed, 2015)	Various sports in the UCF sports dataset
	Movements including shoot, pass, jump, defense and run	Video	KNN with multi-model feature fusion (Pan and Li, 2020)	Basketball
	Referees' movements	Video	A multilayered perceptron neural network (Pecev et al., 2016)	Basketball
	Twelve fine-grained actions using a tennis racket but without the tennis ball	Image datasets	A three-layered LSTM network (Mora et al. 2017),Pre-trained CNN model and multi-layered LSTM with the historical information (Cai et al., 2020)	Tennis
	Actions including free hit, goal, penalty corner and long corner	Video	The pre-trained VGG-16 model (Rangasamy et al.,2020)	Field hockey
	Key actions pf players	Video	A non-uniform quantized HSV space method (Yan and Woźniak, 2022)	Aerobics

(*continued on next page*)

Table 4 (*continued*)

Genres	Focus	Data source	Methodology in Ref.	Sport events
Vision-based pose recognition	Joint player and stick poses	Video	A transfer learning from pre-trained stacked hourglass network for detecting different joint positions on the player (Neher et al., 2018)	Hockey
	Skeleton-based actions	Image datasets	Stacked Hourglass Networks (Newell et al., 2016),An unsupervised model (Su et al., 2019),A shift graph convolutional network (Cheng et al.,2020), A context-aware graph convolutional network (Zhang et al., 2020),DensePose-RCNN ([Güler et al.,2018)a framework that employs modified Inception-ResNet and swarm intelligence-based metaheuristic (Basak et al., 2022)	Various sports in the benchmark datasets
	3D pose estimation	Multiview image dataset	Random forest for classifying the body parts and 3D poses estimated by back-projected by multi-view images (Kazemi et al.,2013)	Football

(Jiang et al., 2009). Compared to single player detection, multiple player detection in team sports is more complicated. Due to the camera capture angle, occlusion is the obstacle for continuous detection of a certain player. To solve this problem, multiple overlapping cameras are used to obtain information from different scene views (Hamid et al., 2010; Direkoğlu et al., 2018).

In terms of extracting hidden high-level features, deep learning is more effective than traditional methods based on handcrafted features (Wang et al., 2018). Convolutional neural networks (CNNs) have been proven to be effective for player detection (Şah and Direkoğlu, 2018). Region-CNN (R-CNN) and its variations, such as Mask R-CNN and Faster R-CNN, are another type of classic network with better performance in different sports, including roller skating (Bu et al., 2017), handball (Pobar and Ivasic-Kos, 2019) and soccer (Liu and Bhanu, 2019). The attention mechanism is also used to improve the detection accuracy in bicycle races, rowing races and football (Symeonidis and Tefas, 2018).

Compared to player detection, ball detection is more difficult, as a ball typically has a smaller size, higher velocity, and a greater variety of moving directions. Since the ball is much smaller than the player, it can be occluded more easily. Additionally, a moving ball may not be captured under high velocity, and its trajectory may be irregular if a player intercepts the ball suddenly or possesses it for a long period (Kamble et al., 2019a; Kamble et al., 2019b). Like player detection, color, size, and shape features are the most common appearance information used to extract ball blobs in ball detection (Chakraborty and Meher, 2013). However, the color of the ball is affected by the weather and illumination conditions, and the detection result is unreliable if the color of the background is similar to that of the ball. Ball size is usually estimated by the number of pixels according to its actual dimension before searching in video frames. The measurement of the ball shape depends on the ball speed and camera frame rate. Therefore, additional information and mathematical methods are applied to ball detection to implement appearance features for better results. For instance, motion information in the middle shot is used to screen football candidates after their positions are extracted based on shape and size features (Pallavi et al., 2008). However, collision and dribbling will cause appearance deformation and unpredictable movement of the ball. Aiming at this problem, information about the edge roundness of ball candidates and a probabilistic method of the local affine invariant region are used to weaken the influence of the above problems on the ball detection accuracy (Leo et al., 2013). Furthermore, distance information about proximity to the expected area, where ball candidates are supposed to be found near the hitting position, is also used for ball detection in tennis (Yu et al., 2004). Gradient directions combined with shape fitting are explored as well (Almajai et al., 2012).

In conclusion, object detection in sports focuses on vision-based player detection and ball detection. This is because vision-based techniques need to extract the target from the frames compared to other sensor techniques. The methods include various traditional methods based on low-level manual features and deep learning methods that can automatically mine semantic information. Traditional object detection algorithms with handcrafted features are less robust, and high-level features are not fully excavated. However, a large amount of training data annotation is needed for network modelling. With limited samples, few-shot learning has the advantage that only a small part of the labelled samples for each target category are used as training data. The small sample detector not only performs better but also has strong versatility. By predicting the weight of the feature, the model can be changed from one dataset to another different dataset (Kang et al., 2018).

3.2. *Vison-based player and ball tracking*

Video-based tracking systems generally use fixed cameras or broadcast videos to capture object movement and obtain the trajectory. Environmental illumination and scene complexity have a great influence on the continuity and accuracy of video-based tracking systems.

Video-based player tracking is considered together with object detection. Object detection recognizes the player or ball in a frame, while object tracking establishes a correspondence of the detected object between adjacent frames and thus forms a continuous trajectory. Detection and tracking are generally preferred to be processed and analysed simultaneously as a step. The Kalman filter (Kim and Kim, 2009), particle filter (Yang and Li, 2017) and data association (Zhou et al., 2015) are classical algorithms that are applied in object tracking. The Kalman filter is used to predict the position in every frame, and it is suitable for tracking straight line motion, slow motion, and constant acceleration of the object without occlusion. Particle filters are more suitable for object tracking with nonconstant velocity and occlusion. Data association is an offline approach and performs well in team sports, such as tennis, volleyball, and soccer. However, it may have problems of misdetection and noisy background. In addition, graph-theory-based methods involve finding a player or ball route by forming a weighted graph. The weight of each edge is measured by the correlation of the joined nodes (Yu et al., 2007). Energy optimization is another approach for more precise description. A conditional random field (CRF)-based model (Wang et al., 2014) and a constrained Bayesian inference model (Wang et al., 2016) are explored for player and ball tracking in soccer and basketball. Recently, online learning has been studied and performed more robustly in occlusion conditions, but computational complexity has increased. In team sports, the trajectory of one player is highly relevant to the movement of other players. The correlations among player movements are used as contextual information to improve the tracking performance of multiple players (Liu et al., 2009; Liu et al., 2013; Zhang et al., 2018).

Video-based ball tracking is studied in both and three dimensions, especially in sports videos. Tracking failure may occur if there is short-term misdetection under heavy occlusion. To solve this problem, overlaps with players and lines are tracked to estimate ball trajectory and movements (Miura et al., 2009). In addition, the position of the ball under occlusion can be predicted by establishing the mathematical equation of the ball's motion trajectory. For example, the path of the ball is regarded as a parabola, and the missing position can be predicted by interpolation (Chakraborty and Meher, 2013). However, the trajectory of the ball cannot always be fitted by a mathematical equation. Considering that trajectory is nonlinear sequence data and difficult to model and predict, a recurrent neural network (RNN) structure is applied for trajectory regression (Shah and Romijnders, 2016). A 3D trajectory is generally reconstructed based on the 2D trajectory using the camera parameters calibrated by multiple views and the physical characteristics of the ball motion. In addition, CNNs can also be used to track the 3D trajectory of a ball, even under high-speed and occluded conditions (Kamble et al., 2019a).

In conclusion, vison-based tracking methods are connective to object detection. Although sports videos and broadcasts are easy to access, illumination and occlusion may lead to tracking loss. Multiple cameras covering the whole court may solve the occlusion problem to some extent. In the case of low illumination, the image does not have enough features for visual processing. Generally, this problem can be alleviated by making additional datasets, but the disadvantage is that it is very labour-intensive. A domain adaptive-based model that can be applied to low light conditions has been studied recently by transfer learning and knowledge distillation (Sasagawa and Nagahara, 2020).

3.3. Vison- based action and pose recognition

Action recognition not only needs to analyze the spatial dependencies of objects but also needs to analyze historical information. Action recognition can be roughly divided into video-based methods and device-based methods according to sensor technologies. In addition, video-based action recognition includes traditional methods with handcrafted feature extraction and deep learning with automatic feature extraction. The process of traditional methods is to first extract features of key points, then cluster the feature descriptors, and finally complete the action recognition through the classifier.

There are two kinds of action recognition. One is general action recognition, which addresses the whole body (Fig. 5), and the other is skeleton-based action, also called pose estimation, which addresses the joint posture of the body (Fig. 6). Pose estimation mainly detects key points of the player, such as joints and facial features.

1) Video-based action recognition

For general action recognition, the space–time interest points method and dense trajectory method are representative traditional methods. The space–time interest points method encodes the video information in spatiotemporal coordinates, and the trajectory method tracks the change in the given coordinate image over time. After key point extraction based on the above methods, histogram of oriented gradient (HOG), histogram of flow , motion boundary histograms and scale-invariant feature transform are the most common feature descriptors. They essentially divide the image into many small cells and count the gradient values in all directions. The combination of feature descriptors and classifiers, such as sparse multinomial logistic regression classifiers and support vector machine classifiers, can be used to recognize the actions of players in hockey (Lu et al., 2009), jumping (Chen et al., 2013) and basketball (Pan and Li, 2020). The trajectory based on key points may be discontinuous due to occlusion, which affects the accuracy of action recognition. To overcome this limitation, a weighted trajectory is used according to the accumulated global distribution of feature points (Yu et al., 2014). The improved method can describe motion information even under occlusions, viewpoint changes, and scale variations. Recent work shows that contextual information can improve the accuracy of recognition, including spatial context (Hasan and Roy-Chowdhury, 2015) and temporal context (Ramanathan et al., 2015). Markov random field (Akbarian et al., 2016) and dynamic graphic model (Xu et al., 2019) were constructed to fully take advantage of context information.

Deep learning is another method that has been widely used for action recognition in the past decade. Two-stream networks (Simonyan and Zisserman, 2014) are an important step in video-based action recognition. In this network, the CNN model is trained for video images and dense optical flow. Various strategies for spatial and temporal information fusion have been studied for better performance (Feichtenhofer et al., 2016; Wang et al., 2016). Based on basic networks, multilayered perceptron neural networks and LSTM are also used to recognize the movement of referees and players in basketball (Pecev et al., 2016) and tennis (Mora and Knottenbelt, 2017). Furthermore, a pretrained VGG-16 model was used and fine-tuned for hockey activity recognition (Rangasamy et al., 2020). Without fine-tuning, the sequences of features from the pretrained model can be used as the input of stacked LSTMs for tennis action recognition. Similarly, in addition to a pretrained CNN model and multilayered LSTM, a history information layer was added to analyze some sports videos (Cai et al., 2020).

2) Video-based pose recognition

Pose recognition, that is, skeleton-based action recognition, is more useful for referees and coaches to evaluate player performance. 2D pose estimation methods can be divided into two categories: top-down approaches and bottom-up approaches. Top-down methods detect the bounding box of the human first, and then skeleton key points are detected from every bounding box. Bottom-up methods detect all joints first and then combine them into a whole body. With the gesture displayed by skeleton key points, player action can be recognized. In top-down methods, because object detection has been expounded on in the above section, key point detection will be highlighted here. Regional multiperson pose estimation and cascaded pyramid networks are the top-down skeleton key point detection models proposed in recent years, and convolutional pose machines and part affinity fields are the bottom-up models. Moreover, stacked hourglass networks (Newell et al., 2016) are a classic model with the ability to capture information at various scales, to position the key points and to realize pose estimation in sports, such as hockey (Neher et al., 2018). In addition to supervised learning, the unsupervised model for skeleton-based action recognition is explored and achieves similar accuracy to supervised learning (Su et al., 2019). Furthermore, graph convolution networks (GCNs) are being studied with increasing interest, as GCN is a local operation. To capture local joint connectivity, multiple scale structures and long-term temporal relations at the same time, a shift GCN was explored for robust recognition (Cheng et al., 2020). In addition, long-distance dependencies between joints were used as context information to generate context-aware GCN (Zhang et al., 2020).

When 3D human body posture is needed, 2D skeleton point detection cannot meet the requirements. Because sports videos and broadcasts are generally two-dimensional RGB images, 3D pose estimation is usually realized by three-dimensional reconstructions. For example, multiple cameras are used to capture multiple views to estimate a player's 3D posture (Kazemi et al., 2013). For example, DensePose-RCNN is

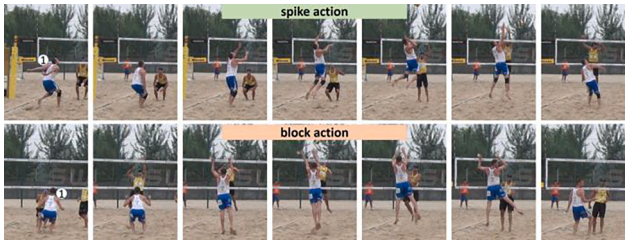


Fig. 5. General action recognition (Kong et al., 2020).



Fig. 6. Skeleton based action recognition (Neher et al., 2018).

proposed to establish the dense correspondence between 2D RGB images and 3D human bodies (Güler et al., 2018).

4. Precision sports analysis based on location and motion tracking technologies

In this section, various application analytics in sports are presented, including kinematics analysis, event detection and prediction, match performance analysis, and visualization analysis applications.

4.1. Kinematics analysis

Kinematic analysis mainly includes temporal motion analysis and dynamic motion analysis, which are based on basic motion measurements, such as velocity, orientation, and position. It is important for training quantity and quality assessment and performance evaluation of players by analysing the kinematics of players and sports equipment.

Temporal motion analysis. Temporal motion is about the beginning and the end of a movement, phase recognition and segmentation, such as stride, step and standing duration, number, and frequency. For instance, to evaluate the performance of elite and recreational players in ice hockey, critical motion parameters are analysed based on high-speed video and accelerometer, including final puck velocity, puck acceleration, blade-puck contact time, maximum bending and angular velocities of stick shaft (Villaseñor et al., 2006). Motion parameters can be regarded as indicators to evaluate the standard degree of movement. For example, peak outwards acceleration is used as a critical indicator of illegal actions assessment in cricket (Spratford et al., 2015). Furthermore, main phase duration detection of various sports can provide valuable performance evaluation for coaches and players. For instance, jumping technical actions are evaluated by measuring the duration of the main stages of ski jumping automatically by sensors mounted on the player, including takeoff release, takeoff, and early flight (Chardonnens et al., 2012). Moreover, the tumble turn of the swimmer, including the approach time, contact time, glide time, kick time and stroke time, are calculated for elite swimming performance analysis (Le Sage et al., 2013a). In addition to tumble turn, time that the back foot leaves the tackle, time that the hip enters the water, time of the first kick, time of the first stroke, the number and rate of kicks underwater, and the number and rate of strokes are also analysed for swimming performance evaluation (Le Sage et al., 2013b).

Dynamic motion analysis. Dynamic motion analysis of players and sports equipment is important to understand the skills and performance of players. The dynamics of the equipment can be determined by the velocity and angular velocity when the player releases it, as well as frictional contact with its surface. For example, the spin dynamics of a bowling ball, including the acceleration and angular velocity, are accurately measured to assess bowler skills (King et al., 2011). In addition to the ball, the punching bag embedded with an accelerometer is used to estimate the punching and kicking force as well as the reaction time of combat players (Buśko et al., 2016). Similarly, the inverse dynamics relationship between the reduction in momentum of different boxer arm segments and the impulse of the impact force are explored to help boxers improve their professional skills (Nakano et al., 2014). For alpine ski racing, in addition to using inverse dynamics, it is also necessary to understand the load ratio between skis. The load ratio between the skis is obtained through the pressure sensor insoles embedded in the ski boots. The purpose is to estimate the ground reaction force, aerodynamic force, snowboard friction, power and energy received by each foot, thereby capturing 3D kinetics and kinematics (Brodie et al., 2008). Furthermore, the joint dynamics of the player are also studied as a dynamic parameter, such as leg swing time, force, torque, angular momentum, angular force and angular impulse, and these dynamic parameters are analysed during the kicking process in football (Meamarbashi and Hossaini, 2010). Body inclination and the center of mass of the player can also be regarded as joint dynamics, such as the complex cyclic motion (Cunniffe et al., 2009) and the displacement of the sacrum (Cross et al., 2015) in ski skating, as well as the velocity and position in alpine ski racing (Gabbett and Domrow, 2007) and the velocity in running (Gallo et al., 2015).

4.2. Event detection and prediction

Events typically consist of actions that occur in chronological order, which contain more semantic information. Basically, broadcast videos are the data source for event detection, from which game highlights can be discovered. Existing methods of event detection can be classified into two categories: one uses video content only, and the other uses additional knowledge in addition to video content. Broadcasts are a special type of video that sometimes includes texture, replays and cheers from audiences, which can be fully used as additional knowledge. For instance, the occurrence of critical events is usually accompanied by cheers or included in the replay.

Sports video content contains audio, visual and text features, which can be extracted by rule-based, statistic-based or machine learning-based methods. Some studies only use a single feature for event detection. For instance, only audio-track features are used for extracting highlight events from baseball videos (Rui et al., 2000), and visual features are used for event detection in soccer (Ekin and Tekalp, 2003) and baseball (Chen et al., 2012). However, to improve the accuracy, multiple features are integrated for event detection in different field sports, including soccer, rugby, hockey, and Gaelic football (Sadlier and O'Connor, 2005). In addition to low-level features detected by classifiers, such as SVMs, mid-level semantic features are explored by Bayesian networks and extreme learning machine-based classifiers for key event detection (Tavassolipour et al., 2014; Zhao et al., 2015; Javed et al., 2019). To extract high-level features, deep learning is also used to detect semantic events in team sports (Wu et al., 2020) and field sports (Kapela et al., 2015).

Event prediction captures the dynamic motion and automatically uses reasoning to predict a complete event sequence. For example, the batting propensity of a player can be predicted based on his or her position on the court and the defender's position. Predictions of events are generally based on specific data. For example, to predict offensive passing, shooting, and dribbling in basketball games, spatial shooting, passing tendency and the duration of possessing the ball are comprehensively considered and analysed by mathematical models in real time (Yue et al., 2014). The probability of scoring a shot can be predicted by extracting the shooting position and angle (Lucey et al., 2015). Moreover, the expected score of ball control in basketball matches is predicted by analysing the position of the ball and the players (Cervone et al., 2014). In tennis, the possible type and position of the next shot can also be predicted based on Hawk-Eye data (Wei et al., 2016). Similarly, event prediction of baseball pitches has been studied (Ganeshapillai and Guttag, 2012). To obtain better performance and robust results, deep learning is also used in event prediction (Sicilia et al., 2019).

Sports event detection and prediction is a crucial task in context understanding and mining. An event usually contains multiple frames, and a sequence of player actions and other objects in the frames are part of the event, which can be used as semantic information for highlight detection and prediction.

4.3. Match performance analysis

Match analysis includes two aspects: individual performance and team tactical analysis. The description of individual performance can be realized by locomotor trajectory and physical characteristics, while team performance and tactic analysis need to obtain meaningful strategic characteristics according to the arrangement and interaction of player movement in the team. It can be believed that both players and teams have their own spatial–temporal patterns and that match performance can be learned from the trajectories of players and balls.

Individual player performance analysis. The performance of the player is generally analysed through historical data. For example, by analysing data for one season, factors affecting the performance of top football players are explored, including seasonal changes, opponent levels and own physical activity (Rampinini et al., 2007). The performance of the basketball player under different scoring modes is also studied, including the way and position of possessing the ball, play configuration and defensive state (Almeida et al., 2016). The performance of players in passing, dribbling, and shooting can also be evaluated by scoring offense and ball control (Cervone et al., 2014), and all-star and nonall-star NBA players are compared by statistical performance data (Sampaio et al., 2015). In addition to analysing players' performance based on statistical characteristics of historical data, neural network-based methods are also widely used. For instance, a network model of the offense is proposed to analyze and predict the skills and performance of the player using tracking data (Skinner and Guy, 2015). Furthermore, a collaborative strategy combining neural networks and mathematical models has been explored for performance evaluation, such as the defensive skills of soccer players (Le et al., 2017).

Team tactical analysis. Team tactical analysis is more complicated than individual player analysis because team sports analysis relates to movements and interactions of players, and movements of the ball are also involved in ball games. Entropy maps are used to represent the predictability of soccer team behaviours and detect team style variation (Lucey et al., 2012). Similarly, a descriptor generated from the entropy of each player is studied to identify team styles (Bialkowski et al., 2014). To describe player patterns and analyze team strategies, the structure of player trajectories is explored based on the hierarchical model (Miller, 2017). Furthermore, neural networks are also applied for team tactic analysis. For instance, RNN is used for the analysis of offensive play by converting trajectory to image (Wang and Zemel, 2016). CNN is also a widely used model to learn discriminative features of team trajectories and obtains good results in sports involving complex player interactions, such as ice hockey and basketball (Mehrasa et al., 2018).

4.4. Sports visualization analysis

Although sports analysis from videos and other sensors can digitalize the performance of the player and the game, sports visualization can provide coaches and referees with a concrete digital picture of player performance to better observe and develop personalized training plans for each player. This kind of immersive mode is mainly realized through VR and AR systems at present.

VR technology uses computer simulation to create a virtual world to make people immersive, and things can be observed in a timely and unlimited way. When the player is moving, complex calculations can be performed immediately, and precise information is sent back to create a virtual world. STRIVR opens a new field of sports training using VR technology. Various players and coaches have used this technology for training. A VR-based training system called KaraKter was developed to help Karate Kumite for players, and the training effect of the system has been approved by experts (Zhang et al., 2018). The VR system can also generate feedback for the trainee by analysing real-time performance. To guide the player more scientifically, a VR training feedback system including predefined messages and visual augmentations is explored to detect the error of the player during exercise (Miranda et al., 2018).

AR is used more widely at present because the function of VR has not been realized, and VR-based products are either too fundamental or unrealistic due to their high cost and complex technology requirements. AR technology can not only fully reflect the content of the real world but also display virtual information content based on the real world. Real environments and virtual objects can coexist in the same 3D space at the same time to help users understand and decide their behaviours (Rogers et al., 2018). In fact, AR technology has long been used in sports competitions. For example, when watching a swimming race on television, the nationality and name of the contestant can be shown in real time on each track. For simulation, a markerless mobile AR system has been explored in basketball applications (Santoso, 2018). In sports training, various AR systems for specific sports are developed to improve the performance of players, such as high precision judo based on an AR training system integrated with Microsoft Kinect (Sieluzycki et al., 2016). AR can also help the coach explain technical actions in sports. For example, a self-calibrating camera-projection system based on an AR system was developed to learn climbing techniques (Wiehr et al., 2016).

4.5. Entertainment and leisure

Sensor technologies for location and motion are not only applied in professional sports but also cover entertainment and leisure in daily life. They are applied in computer games or somatosensory games about sports, which can motivate the physical activity of consumers. Fitness and health monitoring also apply location and motion technologies to help consumers live better.

Gaming. Different from traditional keyboards and mice, the combination of sensor technology with visualization systems can provide a more authentic and immersive experience. Action information of consumers can be captured by sensors and used as the input of VR or AR systems, improving the immersion and realism of the game. In initial work, only sensors were used only in gaming. An interactive game is developed for users to perform an interactive game based on a camera (Höysniemi et al., 2004). With the emergence of visualization systems, interactive simulation sports games based on VR and AR systems have been explored to generate realistic game experiences. Nintendo Switch is a typical representative of gaming systems for entertainment. The popular Pokémon Go is an AR game coproduced by Nintendo, Pokémon Company and Google's Niantic Labs. Consumers can use their smartphones to discover virtual game roles in the real world and capture and fight with them by operating the phone.

Fitness. There are currently many wearable wrist-type devices, such as the HUAWEI band and Fitbit, that can accurately monitor and analyze the user's heart rate, running speed, calorie consumption and sleep quality. In addition to wearables, smartphones have become dominant in consumer portable belongings because the functions of fitness wearables can be replaced by applications on smartphones. These applications can calculate food calories and monitor walking steps and running speed. Keep is a popular fitness application installed in smartphones that realizes outdoor running and cycling functions based on the built-in sensor of the phone. The outdoor running function can record the user's movement trajectory, running distance and speed. The cycling function integrates trajectory, cumulative climbing and calorie consumption.

Health monitoring. Sensor technology is often used to monitor the health of elderly and disabled individuals for remote viewing by guardians and doctors (Bonizzoni et al., 2018). However, the elderly may often forget to wear these devices, so a low-cost intelligent home fall detector based on an artificial vision algorithm is explored (Miguel et al., 2017) and achieves high precision in fall detection. Wearable IMU is applied in injury prevention of overhead sports (Rawashdeh et al., 2016), such as in tracking baseball throws and volleyball serves. For rehabilitation without the constant presence of trained and experienced technical personnel, a swimming device and real-time feedback application is developed to provide precise swimming data (Kos and Umek,

2019), such as stroke time and stroke rotation angle symmetry. Wearable devices in the prevention and rehabilitation of running-related injuries are applied as well (Willy, 2018).

5. Sports data privacy and ethics

Location and motion technologies applied to precise sports has greatly expanded the methods of measuring athletes. In addition to sports technology companies, such as Catapult and STATSPerformance, many technology companies have proposed various solutions based on different sports applications as well, as shown in Table 5. In recent years, the overall sports technology market has also attracted a lot of attention. According to a report by SportsTech X, less than 1 billion dollars was invested in the then-fledgling sports technology sector in 2014, while more than ten times that figure is being pumped into the space on an annual basis today (Carp, 2022). The quantification of athletes by location tracking technology has greatly promoted the intelligence and digitization of sports, and gradually changed the way athletes train, compete and manage their careers. However, the inherent risks cannot be ignored, the most important of which is athlete's privacy violation. In particular, the regulatory capacity of the law on technology is significantly lagging, and many ethical issues related to athlete privacy and data security need to be taken seriously.

5.1. Data privacy and ethics

Athlete data can be divided into two categories: competition data of sports events and individual performance data of athletes. Performance data also includes athlete's location/tracking data such as speed and distance covered, and biometric data such as heart rate, recovery rate, fitness, etc. The quantification of athlete data has exchange value and utilization value (Bai and Bai, 2021). The exchange value is reflected in sports data service providers selling athletes' competition and performance data to professional sports teams, game companies, sports enterprises, etc. The value of utilization is reflected in the improvement of tactical level in competitive sports, the prediction of sports lottery results for related events, and the creation of sports simulation games. Most importantly, tactics formulation, player evaluation and selection, and player injury prevention can be completed based on sports data.

Collecting athlete data for sports analysis can help provide athletes and coaches with more personalized training to improve their athletic abilities, and can also help event organizers improve their seating rates, attract more sponsors, and improve broadcast ratings. However, both wearable sensors and cameras can conduct massive data collection and analysis without consulting athletes. This comprehensive digitization based on location and motion tracking technology forces people to consider whether the intrusion of technology into sports has exceeded the necessary boundaries of athlete privacy (Psychoula et al., 2020). Excessive collection of personal data carries the risk of loss and disclosure, and is vulnerable to improper exploitation or hacking by malicious organizations. For example, a former ball scout for the St. Louis Cardinals hacked into the Houston Astros' roster database and e-mail system in 2016, leaking notes on each player's trade status, bonus information, recent performance and injuries (Wells, 2016). In April 2020, the Tokyo Olympic Organizing Committee was attacked by hackers. Around July 2021, there was another incident where a malicious phishing program tried to phishing staff (Trend Micro, 2021). In 2022, due to security issues on the servers of the International Table Tennis Federation, personal information such as passports and vaccination status of hundreds of professional table tennis players were leaked (Verlaan, 2022).

While the in-depth development of data collection and analysis promotes the digitization and precision of sports, its data privacy risks will also cause damage to athletes. Many athletes feel uncomfortable using location and motion tracking technology to collect their data, especially biometric data, and are concerned that even if the use of biometric data is prohibited, the collection of relevant data can lead to deviations in contract negotiations, which can have a negative impact on their career. While the National Basketball Association collective bargaining agreement has concluded that information obtained from wearable technology cannot be used in contract negotiations (Leung, 2017), other leagues, such as the National Football League, have allowed the use of such data. Statistics and metrics are a major bargaining chip in professional sports contract negotiations. For example, based on performance data from athletes, football clubs underpay players by accusing them of fatigue more easily or declining athletic performance earlier than other players (Garlewicz, 2020). There are also concerns among athletes that the required data collection will gradually interfere with their private affairs (Gale, 2015).

The purpose of applying location and motion tracking technology to competitive sports is to improve the skill level of athletes to win competitions. For example, in the 2007 season, the New England Patriots filmed a defensive signal from their own sidelines for the New York Jets in a game (Bishop, 2007). Throughout the 2017 regular season and playoffs and the 2018 season, the Houston Astros used game feeds from halftime cameras to decode signs from the opposing team (Weese, 2022). In September 2020, it has been revealed that a South Korean go player has defeated his opponent using artificial intelligence technology in the Korean Go world (Tang, 2020).

Due to the high commercial value of sports data, the sports data industry has grown into a multi-billion-dollar industry, which in turn has led to the vigorous development of sports betting. The global sports betting market is expected to increase by about $144 billion from 2021 to 2026 (Technavio, 2022). A new issue that arises is the use of personal data of athletes for sports betting, for example using biometric and location data of athletes to set odds or place bets. This has sparked a backlash from athletes who see it as a serious invasion of privacy. "Project Red Card" is an example of the tension created by the sports betting boom. According to reports, a British organization called Global Sports Data and Technology Group Limited has sued 17 sports betting companies and data processing companies for illegally profiting from player statistics(Ornstein, 2020).

5.2. Data privacy protection technology

In the era of big data, the issue of cyberattacks and data breaches affecting the sports industry. To protect the personal data of athletes collected and analyzed using location and motion tracking technologies, a growing number of policy makers and privacy enforcement agencies are considering incorporating privacy-enhancing technologies into privacy and data protection frameworks. privacy-enhanced computing is one of the top nine strategic technology trends in 2021 released by Gartner, the world's leading information technology research and advisory company (Panetta, 2020). Privacy computing is a joint computing technology by multiple participants. The participants collaborate to conduct joint machine learning and joint analysis on their data without disclosing their own data. It allows the collection, processing, analysis and sharing of data while protecting the confidentiality of personal data.

Federated learning, a distributed machine learning framework, is one of the main methods to realize data privacy protection in privacy computing. Early distributed learning integrated data from different sources for modelling, but the joint training and privacy issues of heterogeneous data have not yet been resolved. Google first tried to use federated learning to solve the privacy problem of personal data on Android and performed joint modelling without sharing data (McMahan et al., 2017). Based on this paradigm, personalized federated learning mechanisms are further studied (Sun, 2022). Compared with traditional machine learning, federated learning achieves data independence; that is, data are always stored locally. A joint model is established through the encrypted exchange of data and model parameters, thus meeting the needs of user privacy protection and data security. In addition, laws and regulations on data privacy protection are becoming more stringent and

Table 5
Commercial solutions for sports analysis.

Solutions	Technology	Sport fields	Data analytics of application
STATSports	GNSS	Outdoor sports	Conditioning levels management
GNSSport	GNSS	Outdoor sports	Heart rate of players monitoring
JOHAN	GNSS	Football, hockey, rugby	player performance analysis
Localino	UWB	All	Time-motion analysis
BeSpoon	UWB	All	Time-motion analysis
DecaWave	UWB	All	Time-motion analysis
Ubisense	UWB	All	Time-motion analysis
Dartfish	Video	All	Player performance identifying to review
Second Spectrum	Video	Basketball	Dynamic visual feature collecting (e.g., movement)
Game Breaker	Video	All	Play back for daily training
WSC Sports	Video	All live sports	Live sports broadcasts analysis
SportVU	Video	Basketball	Dynamic capture, trajectory analysis, data extraction
TRAKUS	Microwave receiver	Ice hockey, golf, horse racing	Providing position, speed, acceleration, and strength
Motion Analysis	Camera	Eurhythmics	3D full-body motion capturing for movement analysis
Qualisys	Camera	Running, baseball, cycling, golf, swimming	Offering human biomechanics
Vicon	Camera	All	Motion capture comprehensive service
OptiTrack	Camera	All	3D tracking data providing
Hawk Eye	Camera	Cricket, tennis	Tracking and displaying the ball's trajectory
GoalControl-4D	Camera	Football, Hockey, Water Polo	Recording and displaying the ball position
ChyronHego	Cameras	soccer	Sports tracking, analysis, and real-time data visualization
Prozone Sports	Cameras	Team sports	Performance analysis
Catapult Sports	IMU + GPS	Football, baseball, basketball, rugby, cricket	Player performance analysis
ShotTracker	IMU + Vision	Basketball	Deliver statistics and data analytics to teams, fans
Triton Wear	IMU + Vision	Swimming	Detect fluctuations or acceleration in the water
TuringSense	IMU	All	Motion analysis of the player's entire body
iSen	IMU	All	Analysis of the human body's motion posture
Aktos-t	IMU	All	Motion capture of human joints, links, and whole body
Motus	IMU	Baseball	Inform users of potential problem
Stancebeam	IMU	Cricket	Bat Swing, bat speed, bat angles power and shot efficiency
Str8bat	IMU	Cricket	Shots, timing, bat angle, bat speed
Zepp	IMU	Baseball, golf, tennis, football	Sports parameters measurement, such as speed, angle, and distance
Kinexon	IMU	Basketball, American football, handball, ice hockey	Precise tracking
Actofit	IMU	Badminton	Tracking the speed and power of shots, playing style analysis
Wisehockey	BLE AOA	Ice hockey	Tracking players and pucks, performance analytics, media entertainment
Quuppa	BLE AOA	All	Judging and evaluating player's movement
Specto athletics	LiDAR	Track and field	Average speed before takeoff (pole vault, long jump, triple jump)

comprehensive, and enterprises need to ensure that data collection must be open and transparent and that data cannot be exchanged without authorization. Past artificial intelligence algorithms become less viable under these strict data privacy protections. Therefore, federated learning with higher security requirements and privacy requirements is required to realize the provision of big data products and services.

Although federated learning improves the generalization ability of models and protects data privacy, there are still certain security risks. The main sources of security vulnerabilities include insecure communication protocols, maliciously modified data and model parameters, damaged central servers, weak aggregation algorithms, and developers who ignore security precautions (Mothukuri et al., 2021). Security attacks affect the accuracy and robustness of federated models by impeding normal model training or inducing incorrect model predictions (Fung et al., 2018; Bhagoji et al., 2019; Bagdasaryan et al., 2020). Data sanitization (Shen and Sanghavi, 2019), anomaly detection (Li, Sahu, Talwalkar, & Smith, 2020), and knowledge distillation (Li and Wang, 2019) are common defense methods to mitigate the security risks of federated learning systems.

Despite the privacy-preserving features of federated learning, studies have found that passing a small amount of model gradient information can also leak raw data (Bos et al., 2014). To solve this problem, secure multiparty computation, homomorphic encryption and differential privacy are used as the main encryption algorithms for federated learning to enhance privacy protection. Secure multiparty computing mainly protects private data by encrypting parameters (Bonawitz et al., 2017). Homomorphic encryption is a strategy that encodes parameters before addition and multiplication operations to solve data leakage (Ilias and Georgios, 2019). Differential privacy aims to protect privacy by adding random noise to the original data (Kharitonov, 2019). There are also studies on combining multiple encryption technologies to generate more secure and accurate models, such as the DPGAN model that applies differential privacy to the GAN model (Xie et al., 2018), the DPFedAvgGAN model that combines the generation model, federated learning, and differential privacy (Augenstein et al., 2019), and an integrated solution that combines secure multiparty computing with lightweight differential privacy (Hao et al., 2019).

There are more than 2900 open-source federal learning projects on github, including Google TTF, OpenMind Community PySyft, Intel OpenFL, WeChat FATE, and ByteDance FedLearner. Federated learning has been applied in fields such as finance (Yang et al., 2019), smart cities (Gu et al., 2020, Rasha et al., 2023), health care (Brisimi et al., 2018; Pfohl et al., 2019) and smart homes (Aïvodji et al., 2019; Yu et al., 2020). Considering that the current sports field is also facing data privacy and security issues in the era of big data (Chang et al., 2019), there have also been some attempts to use federated learning to protect the privacy of sports data (Karakaya et al., 2022). With the rise of federated learning and the increasing awareness of data privacy protection, federated learning will also have broad application prospects in the field of sports.

6. Challenge and future direction

A comprehensive survey of existing location and motion technologies is conducted in the sports field. Although various advanced positioning technologies have been applied in both outdoor and indoor sports, there are still some challenges in using these technologies for digital precise sports applications and data analytics.

6.1. Location and motion sensor technology development

Although sensor-based and video-based technologies are widely used for location and motion, the hardware devices of various technologies need to be further developed to be able to measure highly dynamic irregular motion with high precision.

The challenge of technologies based on sensors is the accuracy and reliability of sensor measurements without interfering with the environment and the player. In addition to sports bracelets with low-cost sensors for fitness, the location and motion of players and sports equipment are measured by embedded high-precision portable sensors. With the miniaturization of electronics, it has become possible to use GNSS and miniature IMU in various sports. However, unpredictable multidirectional changes can reduce the accuracy of positioning and tracking, even with high-precision sensors, for motion with intense physical conflict. Therefore, attention should be given to the influence of movement speed, nonlinear movement and signal interference on the location and motion results when these sensor-based techniques are applied to precision sports evaluation. In addition, due to the requirement of competitive sports, players must compete physical activities at high speed, high against sports, and any external factors may interfere with performance. Therefore, the integration and miniaturization of sensors are particularly important, and data interconnection should be further improved.

The challenge of technology based on cameras is to obtain an appropriate video sequence for reliable detection, recognition and tracking in a complex environment containing multiple objects. Although this goal is currently being developed using various methods, the quality of videos, relative size of the target object in the frame, lighting changes and frequency of occlusion make the application of video to the sports environment more complicated. Moreover, they are limited in capture volume, allowing the analysis of only a short sequence. However, significant progress has been made in fully automatic motion tracking and reconstruction. The advances mainly include tracking multiple people in outdoor scenes, reconstructing human motion from multiple views, fine-grained posture estimation in natural scenes, analysis of human behaviors and movement and understanding of game tactics.

Recently, light detection and ranging (LiDAR) technology has started to be used in tracking the motion and velocity of players. The LiDAR sensor placed on a tripod is placed at waist height for running speed measurements, and due to the wide scanning angle, the average speed of the player can be obtained. In such an approach, the center of gravity of the runners needs to be used to correct the distance measurements. Furthermore, several 3D-LiDAR were used for player tracking at a soccer stadium. Data on each player's running speed and distance could be gathered, and a heatmap of the player's activity could be created. An interesting new sensor technology is Flash LiDAR. As it is capable of recording 3D images of a scene at more than 200 frames per second, it is ideally suited to be used in real-time applications. The disadvantages of Flash systems include limited image size, resolution, and limited range. Autonomous driving requires mapping of the dynamic and static objects at a rate of several dozen Hz. Such sensors are highly usable for sports since the requirements for sports sensors are similar to those of autonomous driving sensors. Therefore, the development of autonomous driving sensors will eventually lead to the development of location and motion tracking technologies in sports.

Future developments will focus on body parts detectors integrating more sensors, lightweight smart clothing and shoes with no interference with players and scenes, which will improve the reliability of motion location and motion tracking in chaotic environments. Another area of focus is the reliability selection of people and behaviors from low-quality sensor data or images. In addition, further research is needed to improve the processing speed of automatic location and tracking while simultaneously reducing computing cost and maintaining processing power.

6.2. Deeper semantic information extraction and integration

The development and innovation of various sensor technologies make more data available for analysis, but traditional methods are unable to extract sufficiently hidden context and knowledge. More data from a variety of emerging sensors are helpful for precision performance evaluation and decision-making in training and competitions of sports. Big data analysis enables the discovery of potential semantic concepts. Therefore, a better understanding of game tactics and performance may be achieved by exploring more comprehensive theoretical analysis methods and models based on big data (Rein and Memmert, 2016). The advanced development of machine learning and data mining has led to the solution of large-scale problems in the real world. Data mining displays the interrelationship of various characteristics, thereby discovering certain sports patterns. Machine learning is widely used in big data processing and enables sports data analysis to be diversified and large scale, but handcrafted features used in machine learning methods are not sufficient to fully extract the knowledge in sports analysis. The emergence of deep learning with high-level feature extraction makes data-driven methods more intelligent for large-scale data analysis.

Deep learning methods have been proven successful in sports analysis. For example, CNN has advantages in high-level semantic modelling, and RNN is advanced in temporal dynamic modelling. Since different sports have domain-specific semantic concepts and characteristics, the challenge of bridging the semantic gap between sports events has yet to be overcome. In short, the training and test data may come from different domains, so how to solve the problem of transfer learning is still challenging. Learning information from different sports is a promising research direction. In addition, it remains to be studied whether the tactical analysis of different sports can be represented by the same model. A scalable research direction is to develop robust feature extraction solutions to understand motion details in sports. Recently, some contributions have focused on extracting precise object trajectories using spatiotemporal neural networks to improve accuracy and robustness. Therefore, intelligent systems need to be developed to determine the relationship between trajectories and actions and perform higher-level tactical analysis.

Deeper context learning on a large scale remains to be studied. Although there have been successful applications of trajectory tracking and action recognition methods, as well as event prediction and tactical analysis, there are still many problems to be solved due to the diversity of sports. In the future, there is room for artificial intelligence in sports tactical analysis. Especially in team sports, more powerful features need to be explored for robust movement and behavior analysis of multiple players. High-level semantic information must be combined with low-level and middle-level representations for sports style and tactics understanding. Furthermore, multiple devices of different technologies integrated in sportswear in combination with other technologies may be able to provide more abundant data to improve the analysis level of precision sports in the future (Wang et al., 2019).

Further evaluation of network performance is needed to determine the generalization ability of deep learning in sports, such as finding more semantically similar movement patterns in different sports. In addition, one of the major challenges of deep learning analysis systems is to provide many meaningful and differentiated data so that there are enough training samples. Only in this way can deep learning bring precision motion to a new level of reliability and accuracy. Furthermore,

large-scale data processing methods, cloud computing and big data analysis will jointly ensure the reliability and authenticity of the dataset. The collected data will also be more effective and nonredundant, and the various indicators are relatively unified and can be compared with each other. The transmission, analysis and sharing of sports data between researchers and coaches can be realized more easily through cloud interconnection; thus, more scientific and effective training plans can eventually be formulated.

6.3. Future personalized sports and immersive visualization

Although some systems are available for commercial and professional use in sports, there are still several issues need to be addressed, including tracking, display, latency, and content, ergonomics, and design. When using VR to watch sports activities, the user's ability to interact with others is limited. The use of AR technology can provide additional information and superimposed in the real scene, enhance the user's understanding and knowledge, help users to check their actions, to improve their motor skills. In addition, it may be helpful for improving training performance of players and guidance of coaches by enhancing the diversity of systems and allowing personalized content. In addition, it can improve the spectators' experience of watching sports events.

Future systems should be lighter, more integrated, and more stable. Considering the lack of interactivity of VR systems, AR is more promising to further improve the sports visualization, since more information can be exchanged through voice and gestures. Organizers of sporting events are integrating sensors, cameras, and digital signage to attract visitors. In addition, if you want sports to play a better role through AR or VR, players should be guided to use these media to expand their market applications. It can also be further studied and explored to add more elements, such as increasing the playback function.

6.4. Privacy protection and legislation enhancement

Society has increasingly attached importance to privacy protection, and privacy technology has gradually been applied to data protection, but currently there are still some technical difficulties. Privacy protection federated learning has high computational efficiency and communication costs. It typically requires the data transfer between multiple devices, and technologies such as multi-party secure computing for privacy protection can significantly increase the burden of computing and communication. Therefore, it is necessary to research more efficient security algorithms and protocols to reduce computing and communication costs. In addition, federated learning affects computational accuracy or accuracy. Since differential privacy technology can add noise to the data, which may lead to a decrease in the accuracy of the system's calculation results, it is necessary to design more accurate noise processing algorithms to ensure calculation accuracy and accuracy. Currently, there are some studies that combine model compression with privacy protection technology to improve the performance of federated learning (Hamer et al., 2020; Xu et al., 2022; Zhu et al., 2023). In addition, personalized privacy protection schemes need to be designed based on specific application scenarios. Technology is certainly a key means to achieve data privacy protection, but a reasonable and scientific system is also an essential part of the data protection process. Therefore, while accelerating the evolution and application expansion of privacy computing technology, relevant systems such as laws and policies also need to be improved to jointly achieve data protection.

Currently, various countries and regions have introduced corresponding laws and regulations for data privacy protection, such as the General Data Protection Regulation issued by the European Union in 2018, and the Personal Information Protection Law of the People's Republic of China issued and implemented by China in 2021. However, these laws provide a general reference for athletes' data processing and cannot meet the personalized and professional development needs of the sports industry. Some organizations have established rules that clarify how player data is collected, whether it is collected and used voluntarily, how data is protected, and the scope of use. For example, the National Basketball Association of the United States has formed a wearables committee to review and develop safety standards for the environment in which wearable devices are used, and to declare that players have full access to all data collected by approved wearable devices. Although these agreements protect the personal privacy and security of athletes to a certain extent, the terms of such agreements may not be able to keep pace with the rapid development of technological innovation and business development. It cannot be ignored that the unequal relationship between athletes and their data controllers makes it more likely that athletes' personal data rights and interests will be infringed, and the adverse consequences will be more difficult to remedy. Therefore, refining data protection specifications, improving athlete data privacy protection strategies, and strengthening specialized sports industry legislation can ensure the healthy and sustainable development of precision sports based on location and motion tracking technology.

7. Conclusion

A survey of wearable sensor and vison technologies concerning location and motion tracking for movements analysis of human and equipment in sports has been presented in this paper. Related technologies, methodologies and applications have been discussed in detail. The survey shows that there are lots of work ongoing in the precision sports with the development of sensor technologies, especially for location and motion tracking. The combination of location and motion tracking technologies with sports can bring great energy to realize the transformation from traditional sports with several statistic results to precision sports with the whole sports process digitalization. While applying location and tracking technology to the sports field, the protection of athletes' data privacy has gradually attracted attention. However, there are still a lot of issues to be addressed for precision sports industry applicable in coaches, players, spectators and customers. The key issues include the hardware development of technologies, high semantic approaches exploration, changes of training and viewing ways of precision personalized sports and data privacy and ethic issues. Combined with privacy computing technology, cloud computing and big data mining, the integration of technologies and methodologies are the future tendency for the precision sports. At the same time, personal privacy protection and legislation enhancement need to be further improved and perfected.

CRediT authorship contribution statement

Jingbin Liu: Conceptualization, Methodology, Writing – review & editing. **Gege Huang:** Methodology, Writing – original draft, Writing – review & editing, Data curation. **Juha Hyyppä:** Writing – review & editing. **Jing Li:** Resources. **Xiaodong Gong:** Data curation. **Xiaofan Jiang:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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